

Selenium End-to-End Testing — Homework Report

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1. Site chosen and why

Site: <https://www.saucedemo.com> — a demo shop built specifically for practicing Selenium automation. It was chosen because it gives every feature the brief asks for in one small flow: a login form with both a valid and several deliberately-broken accounts (`standard_user` , `locked_out_user` , `problem_user` , `performance_glitch_user`), a product catalogue, an "Add to cart" state change with a visible badge counter, and a cart page — enough for a positive path, a negative path, and a data-driven test without needing to stitch together unrelated pages.

2. Test case table

ID	Action	Input	Expected result	Actual result	Pass/Fail
T1	Open the site (smoke test)	—	Page title is "Swag Labs" and the login logo is visible	Title was "Swag Labs"; logo was visible	Pass
T3a	Log in with valid credentials	<code>standard_user</code> / <code>secret_sauce</code>	Redirect to <code>/inventory.html</code> , inventory container present, 6 products listed	Redirected to <code>/inventory.html</code> ; container present; 6 products listed	Pass
T3b	Add a product to the cart	Add "Sauce Labs Backpack" as <code>standard_user</code>	Cart badge changes from 0 to 1; cart page shows 1 line item named "Sauce Labs Backpack"	Badge went 0 → 1; cart page showed exactly 1 item, name matched	Pass
T4a	Log in with wrong password	<code>standard_user</code> / <code>definitely_wrong</code>	Error banner containing "do not match"; no navigation to <code>inventory.html</code>	Error text: "Epic sadface: Username and password do not match any user in this service"; stayed on login page	Pass
T4b	Log in with a locked-out account	<code>locked_out_user</code> / <code>secret_sauce</code>	Error banner mentioning "locked out"	Error text: "Epic sadface: Sorry, this user has been locked out."	Pass
T6-P2	Log in with empty username	<code>""</code> / <code>secret_sauce</code>	Error contains "Username is required"	Error text: "Epic sadface: Username is required"	Pass
T6-P3	Log in with empty password	<code>standard_user</code> / <code>""</code>	Error contains "Password is required"	Error text: "Epic sadface: Password is required"	Pass
T6-P4	Log in with wrong password (partition duplicate of T4a, exercised via the data-driven test)	<code>standard_user</code> / <code>wrong_password</code>	Error contains "do not match"	Error text matched	Pass
T6-P5	Log in with a locked account (partition duplicate of T4b)	<code>locked_out_user</code> / <code>secret_sauce</code>	Error contains "locked out"	Error text matched	Pass

3. Design technique used for T6

Technique: Equivalence Partitioning (EP) applied to the login form's two inputs (username, password).

Partitions identified for the (username, password) pair:

- **P1 — valid / valid** (both fields correct): exercised in T3a, not repeated in the parameterized test to avoid duplicating the positive-path assertion.
- **P2 — empty username**: `username=""` , `password=secret_sauce` → expects "Username is required".
- **P3 — empty password**: `username=standard_user` , `password=""` → expects "Password is required".
- **P4 — valid username / invalid password** : `username=standard_user` , `password=wrong_password` → expects "do not match".
- **P5 — valid credentials on a disabled account** : `username=locked_out_user` , `password=secret_sauce` → expects "locked out".

These four invalid partitions (P2–P5) are covered by a single `@ParameterizedTest` with `@CsvSource` in `LoginDataDrivenTest` , each asserting the specific error-message fragment that partition should produce.

4. Defects and odd behaviour found

- **Duplicate DOM id on the real inventory page.** `inventory.html` renders two nested elements that both carry `id="inventory_container"` . A strict `findElements(...).size() == 1` check fails against the live site even though the page loaded correctly; the Page Object was written to check "at least one match" instead of exactly one.
- **problem_user — broken product images.** Steps: log in as `problem_user` / `secret_sauce` . Every one of the six products on the inventory page displays the identical placeholder photo (a dog with a tennis ball) instead of its own product image, unlike `standard_user` where each product has a distinct picture.
- **problem_user — "Add to cart" does not update state.** Steps: as `problem_user` , click "Add to cart" on any product. The button stays labelled "Add to cart" and no cart badge appears, whereas the same action for `standard_user` flips the button to "Remove" and shows a badge count of 1.
- **performance_glitch_user — slow login.** Steps: log in as `performance_glitch_user` / `secret_sauce` . There is a clearly noticeable multi-second delay between submitting the login form and the inventory page rendering, compared to a near-instant transition for `standard_user` . Once rendered, the product data itself is correct (unlike the `problem_user` case above).

5. Evidence of a run

Console output of `mvn clean test` (Selenium's WARNING-level DevTools logging suppressed to keep the result visible):

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running et.edu.aau.selenium.tests.LoginPositiveTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 4.786 s -- in et.edu.aau.selenium.tests.LoginPositiveTest
[INFO] Running et.edu.aau.selenium.tests.LoginDataDrivenTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 7.296 s -- in et.edu.aau.selenium.tests.LoginDataDrivenTest
[INFO] Running et.edu.aau.selenium.tests.LoginNegativeTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 3.581 s -- in et.edu.aau.selenium.tests.LoginNegativeTest
[INFO] Running et.edu.aau.selenium.tests.SmokeTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.429 s -- in et.edu.aau.selenium.tests.SmokeTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 9, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] BUILD SUCCESS
[INFO] -----
```